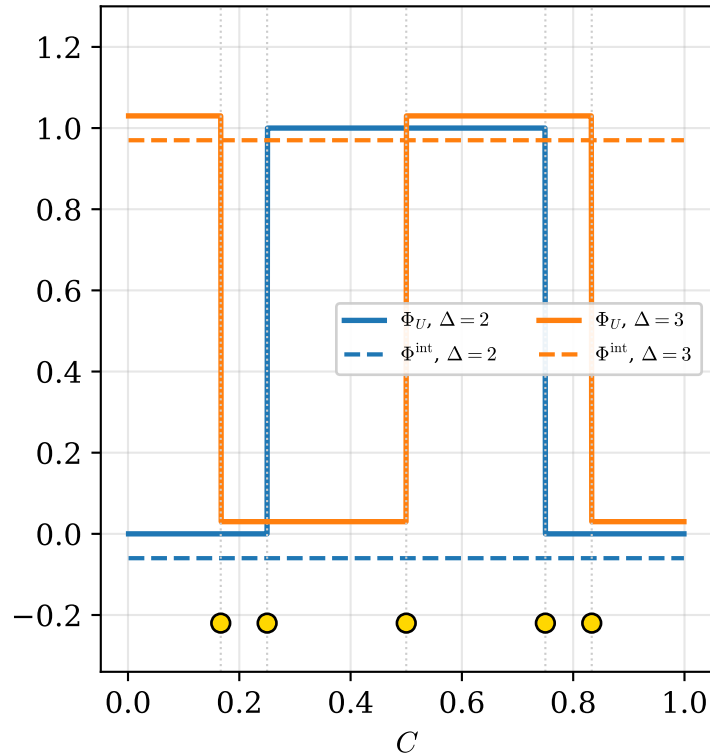
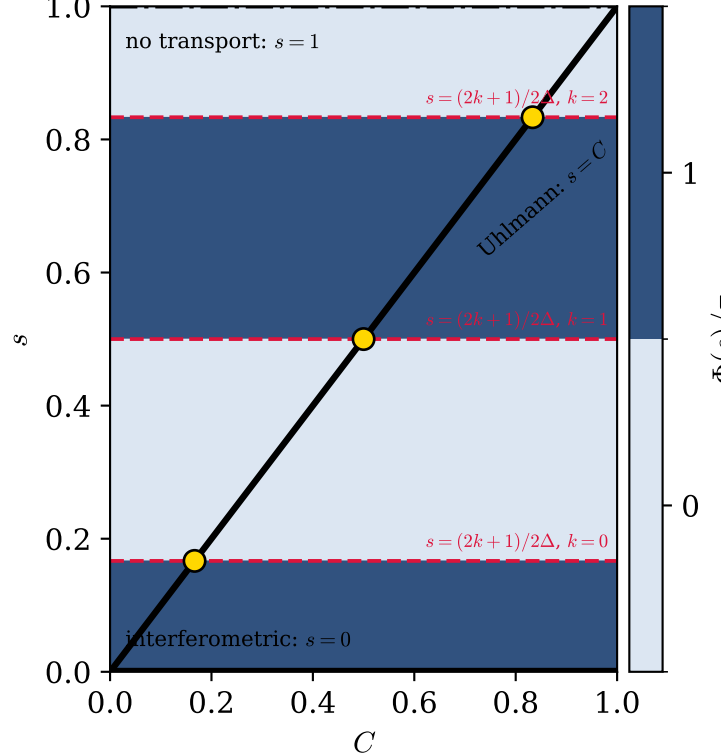


Two mixed-state geometric phases on the same loop: Uhlmann keeps G_d and rescales G_{od} by C ; Sjoqvist deletes G_d and keeps G_{od}

(a) on $p = 1/2$: Uhlmann is a Z_2 staircase, Φ^{int} is a constant parity, blind to C



(b) $G_{\text{eff}} = G_d + s G_{od}$, $p = 1/2$, $\Delta = 3$: $b_s = \Delta s$, nodes only on $s = (2k + 1)/2\Delta$



(c) $\Delta = 2$: $|Z^{\text{int}}| \geq \sqrt{1 - C^2} > 0$ everywhere; black: $|Z_U| \rightarrow 0$ (Uhlmann nodes, gold)

